

THE BIOTA OF INTERMITTENT RIVERS AND EPHEMERAL STREAMS: AQUATIC INVERTEBRATES

4.3

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IN A NUTSHELL

- Intermittent rivers and ephemeral streams (IRES) are temporally dynamic ecosystems that can support a diverse and distinctive aquatic invertebrate fauna
- Resistance and resilience mechanisms allow species and communities to persist in IRES during dry phases and to recolonize quickly once flow returns
- Human influences including climate change and water abstraction alter natural patterns of flow intermittence, with increasing intermittence typically reducing aquatic invertebrate diversity
- Effective management, restoration, and legislation are needed to safeguard the diversity of IRES invertebrate communities

4.3.1 INTRODUCTION

Aquatic invertebrates are a diverse group of organisms that inhabit IRES throughout the world, in regions spanning alpine, arid, Mediterranean, polar, temperate, and tropical climates. IRES invertebrates include benthic, planktonic, and stygobitic taxa (those associated with surface sediments, open water, and groundwater, respectively) and range from widespread taxa to endemic IRES specialists. Classified by size as either macroinvertebrates (those visible without magnification) or smaller meiofauna, macroinvertebrates are more widely studied (Robertson et al., 2000), although meiofauna can also be diverse and abundant in IRES communities (Meyer et al., 2007).

The roles that IRES invertebrates play in ecosystem functioning are strongly controlled by flow intermittence. For example, aquatic invertebrates are important leaf litter processors, but breakdown rates can be reduced in IRES compared to perennial rivers due to fewer “shredders” that feed on such

coarse particulate organic material (Richardson, 1990), reducing the release of finer particulate organic material consumed by other feeding groups. Aquatic invertebrates provide prey for IRES predators such as fish (Courtwright and May, 2013; Chapter 4.5) and are also consumed by terrestrial predators including insects, reptiles, birds, and mammals (Leigh et al., 2013), particularly when stranded in isolated pools or on drying channels. In addition, adult insects emerging from IRES transport aquatic material to terrestrial habitats, providing an important energy source for riparian predators (Lynch et al., 2002; Progar and Moldenke, 2002; Chapters 4.4, 4.6, and 4.7).

Research into IRES invertebrates grows annually (Leigh et al., 2016a), and their diversity and ubiquity make them ideal organisms for addressing many research and management challenges in IRES (Datry et al., 2014a). This chapter reviews aquatic invertebrates in IRES, describing the ecosystems they inhabit, examining how their taxonomic and functional diversity change in response to varying environmental conditions, and considering the adaptations that promote survival in these dynamic ecosystems. We then highlight human activities that threaten aquatic invertebrate communities, explore ways to protect them, and suggest priorities for future research.

4.3.2 IRES AS HABITATS FOR AQUATIC INVERTEBRATES

IRES are complex, dynamic, and diverse ecosystems (Chapters 2.1 and 2.3) in which the shifting habitat mosaics (*sensu* Datry et al., 2016a) associated with intermittence drive the composition, abundance, and diversity of aquatic invertebrate communities (Datry et al., 2014b; Leigh and Datry, 2016; Sections 4.3.3 and 4.3.4). Aquatic invertebrates inhabiting IRES experience highly variable environmental conditions, and critical thresholds at which habitats change profoundly are crossed as IRES transition between flowing, pool, and dry phases (Boulton, 2003). During drying, lateral connections between the main channel and riparian habitats are severed first, then flow ceases and lentic (standing water) pools form. Longitudinal (upstream-downstream) connectivity can be lost, disconnecting pools that may then contract over time. Pools typically have high and variable water temperatures, increasing nutrient concentrations and declining oxygen availability (Lake, 2011, Chapters 3.1 and 3.2). Contracting pools are also characterized by intensifying biotic interactions including predation and competition for limited resources (Boulton and Lake, 1992; Jackson et al., 1999). Ultimately, contraction of disconnected pools may lead to complete surface water loss.

Where flowing surface water is lost, invertebrates may persist in refuges: habitats in which their survival is enhanced (Section 4.3.5). Perennial flowing-water refuges include reaches upstream and downstream of an intermittent reach as well as nearby rivers (Fig. 4.3.1; Boulton, 1989; Chester and Robson, 2011). Persistent pools and other perennial lentic waters are also important refuges (Paltridge et al., 1997), and moist microhabitats including leaf litter, algal mats, woody debris, and damp sediment beneath large stones provide localized refuges in surface channels (Fig. 4.3.1; Boulton, 1989; Robson et al., 2011). Below the drying benthic zone, the saturated subsurface sediments of the hyporheic zone may provide an extensive refuge, particularly when cool, well-oxygenated water fills spacious interstices (Stubbington, 2012). These refuges allow aquatic invertebrates to recolonize intermittent reaches after flow resumes (Datry et al., 2016b; Section 4.3.5).

4.3.3 TAXONOMIC DIVERSITY OF IRES INVERTEBRATE COMMUNITIES

Most studies, typically conducted at the reach scale, have indicated that alpha diversity (defined in Box 4.3.1) declines with increasing flow intermittence, and that communities at sites with greater

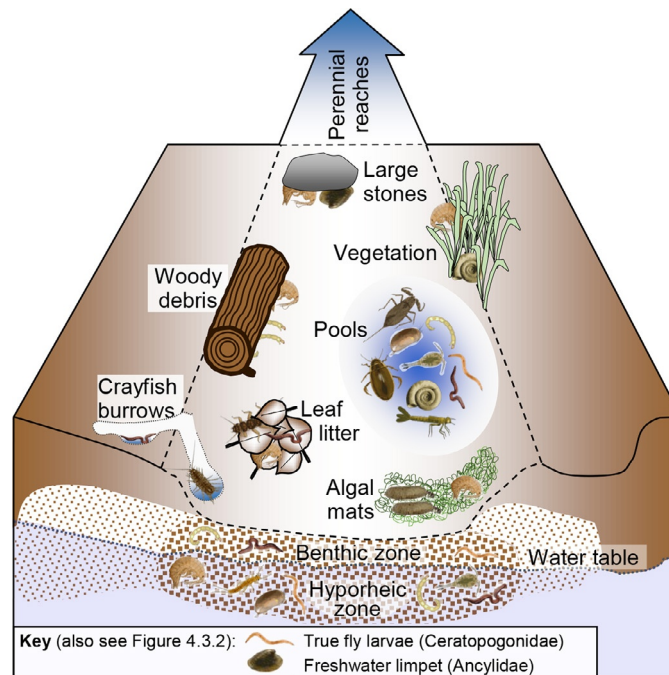


FIG. 4.3.1

In-stream refuges in a conceptualized IRES. Placement of example taxa in refuges is informed by Boulton (1989) and Chester and Robson (2011); other taxa also occur.

Adapted from Stubbington, R., 2012. The hyporheic zone as an invertebrate refuge: a review of variability in space, time, taxa and behaviour. Marine Freshw. Res. 63, 293-311, with permission from CSIRO Publishing. Invertebrate drawings:

Courtesy: Pau Fortuño Estrada. Not to scale.

intermittence are highly nested (Box 4.3.1), comprising subsets of taxa found at less intermittent or perennial sites (Box 4.3.2, Fig. 4.3.2A; Datry et al., 2014b). A comprehensive metaanalysis of perennial river-IRES comparisons revealed that alpha diversity was significantly lower in IRES than in perennial rivers, regardless of climate, season, sampling methodology, or the habitats or river sections sampled (Soria et al., 2017). However, there are exceptions to this pattern: Bonada et al. (2007a) recorded comparable richness at perennial and intermittent sites in a Mediterranean river network, and Dieterich and Anderson (2000) attributed higher richness at forested IRES compared to perennial head-water streams to greater within-reach environmental heterogeneity.

TEMPORAL VARIABILITY IN TAXONOMIC DIVERSITY

The typical decline in richness with increasing intermittence (as characterized by comparing communities present during flowing phases; Datry et al., 2014b) reflects the loss of taxa at critical thresholds as drying proceeds (Fig. 4.3.4; Boulton, 2003; Section 4.3.2). As water levels decline, taxa associated with marginal vegetation are lost first, such as Hydropsychidae caddisfly larvae that feed by piercing aquatic plant cells, and Sisyridae spongilla-fly larvae found on sponge-encrusted riparian tree roots (Boulton and Lake, 2008; Bogan et al., 2015). As flow velocities fall, rheophilic (flow-loving) taxa with high

BOX 4.3.1 DEFINING DIVERSITY: TAXONOMIC, PHYLOGENETIC, AND FUNCTIONAL APPROACHES

Taxonomic diversity refers to the richness (number of taxa) and evenness (relative abundance of taxa) of a community and comprises alpha, beta, and gamma components. Alpha diversity is the local diversity within a community present in particular habitat. Beta diversity refers to differences in taxonomic composition among habitats and over time. This measure includes nestedness (which indicates that at sites with fewer taxa, those present are subsets of taxa at richer sites; [Box 4.3.2](#); [Fig. 4.3.2A](#)) and turnover (replacement of some taxa by others, for example, due to environmental changes; [Fig. 4.3.2B](#)). Gamma diversity is determined by alpha and beta components and indicates the combined diversity across all habitats in a landscape.

A limitation of taxonomic diversity measures is that assemblages with comparable richness and evenness may comprise species of variable relatedness. To remedy this, taxonomic diversity can be used alongside phylogenetic diversity (which considers evolutionary relationships among taxa; [Chapter 4.10](#)) to determine the taxonomic distinctness of a community ([Warwick and Clarke, 1995](#)).

Functional diversity links taxonomic diversity to ecosystem functions provided by a community ([Chapter 4.10](#)) and complements measures of taxonomic and phylogenetic diversity by enabling comparison of taxonomically distinct but functionally similar communities ([Boersma et al., 2016](#)). The functional diversity of a community refers to the range of functional traits possessed by its taxa, where functional traits are characteristics that influence an organism's response to the environment and/or its effects on an ecosystem ([Díaz and Cabido, 2001](#); [Cadotte et al., 2011](#); [Chapter 4.9](#)). Traits include body size, feeding mode, dispersal mechanism, and desiccation-tolerant life stages, and each includes multiple states (e.g., “low” or “high” dispersal, [Poff et al., 2006](#)). Invertebrate trait databases for Europe ([Tachet et al., 2002](#)) and North America ([Poff et al., 2006](#)) have facilitated exploration of the functional diversity of IRES aquatic invertebrate communities ([Vander Vorste et al., 2016a](#); [Leigh et al., 2016b](#)), including regional comparisons ([Statzner and Bêche, 2010](#)), although a lack of trait-related information can necessitate exclusion of meiofauna from analyses ([Boersma et al., 2014](#)).

BOX 4.3.2 NESTEDNESS AND TURNOVER INFLUENCE AQUATIC INVERTEBRATE DIVERSITY IN IRES

The typical decline in alpha diversity with increasing flow intermittence, as characterized by comparison of flowing-phase communities, reflects community nestedness ([Box 4.3.1](#); [Fig. 4.3.2A](#)), with taxa at more intermittent sites (group C) being subsets of those at less intermittent (group B) and perennial sites (group A; [Datry et al., 2014a](#), but see [Bogan et al., 2013](#)). Although variation occurs within each taxon and among systems, perennial sites include a wide range of taxa, such as many mayfly and stonefly nymphs, some caddisfly larvae, riffle beetles, amphipod and isopod crustaceans, true fly larvae, oligochaete worms, and copepod and ostracod microcrustaceans. As flow intermittence increases, rheophilic (flow-loving) and desiccation-sensitive taxa progressively disappear (from group A to B to C, [Fig. 4.3.2A](#); [Datry et al., 2014a](#); [Section 4.3.3](#)).

Despite lower flowing-phase alpha diversity, temporal variation in IRES habitats provides opportunities for taxon turnover ([Box 4.3.1](#)), with different taxa thriving at different times ([Fig. 4.3.2B](#), [Fig. 4.3.3](#); [Bêche et al., 2006](#); [Bonada et al., 2007a](#)). During flowing phases, resilient lotic taxa (group B, [Fig. 4.3.2A and B](#)) dominate and may be joined by perennial stream (group A) taxa. When flow ceases, these taxa are replaced by macroinvertebrates characteristic of lentic pools, including damselfly nymphs, diving beetles, true bugs, and snails ([Fig. 4.3.3](#); [Bonada et al., 2006, 2008](#)), and a meiofauna assemblage including rotifers, copepods, nematodes, and ostracods (Group D, [Fig. 4.3.2B](#); [Smith et al., 2001](#); [Storey and Quinn, 2008](#)). Other taxa, including many true fly larvae and Oligochaeta and certain caddisfly larvae, are generalists that can persist in lotic and lentic conditions (group E, [Fig. 4.3.2B](#)). This taxonomic “time-sharing” ([Bogan and Lytle, 2007](#)) increases the overall, year-round diversity (i.e., temporal beta diversity) ([Bogan et al., 2015](#)) of IRES. Beta diversity increases further when aquatic and terrestrial contributions are combined to estimate total IRES invertebrate richness ([Corti and Datry, 2016](#); [Chapter 4.4](#)).

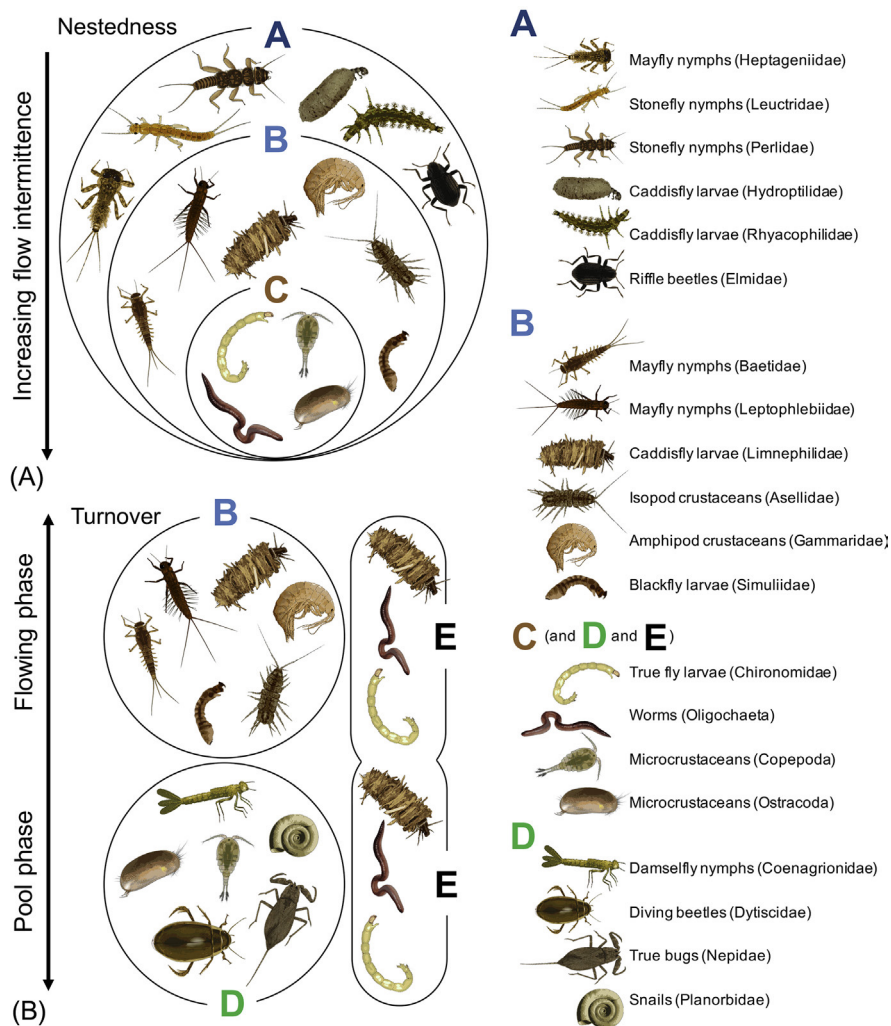


FIG. 4.3.2

Flowing-phase invertebrate community richness typically declines with increasing flow intermittence, with communities at more intermittent sites being nested subsets of those present at sites with greater flow permanence (A). Invertebrate communities exhibit turnover as in-stream conditions transition between flowing and pool phases (B).

Invertebrate drawings: Courtesy: Pau Fortuño Estrada. Not to scale.

oxygen requirements lose their preferred habitat and most are eliminated after flow ceases, including many species of EPT (Ephemeroptera, Plecoptera, and Trichoptera, i.e., mayfly nymphs, stonefly nymphs, and caddisfly larvae; Fig. 4.3.4; Boulton, 2003; Stubbington et al., 2009a). Filter feeders such as Simuliidae blackfly larvae require flow to feed (by filtering fine particles from moving water) and are often also eliminated by flow cessation. In contracting pools, taxa risk elimination if water quality becomes intolerable (Chapter 3.1) or if finite prey resources are consumed (Fig. 4.3.4; Boersma

**FIG. 4.3.3**

Aquatic invertebrates in each IRES phase: flowing phases feature flow-loving caddisfly larvae (a), stonefly (b), and mayfly nymphs (c); the pool phase is characterized by damselfly nymphs (d), true bugs (e), and beetles (f); in dry phases, mussels, cased caddisfly larvae (g) and snails (h) may persist if sediments remain moist; when flow resumes, blackfly larvae (i), dobsonfly larvae (j) and stonefly nymphs (k) are early colonizers.

Photos: Courtesy M. Bogan.

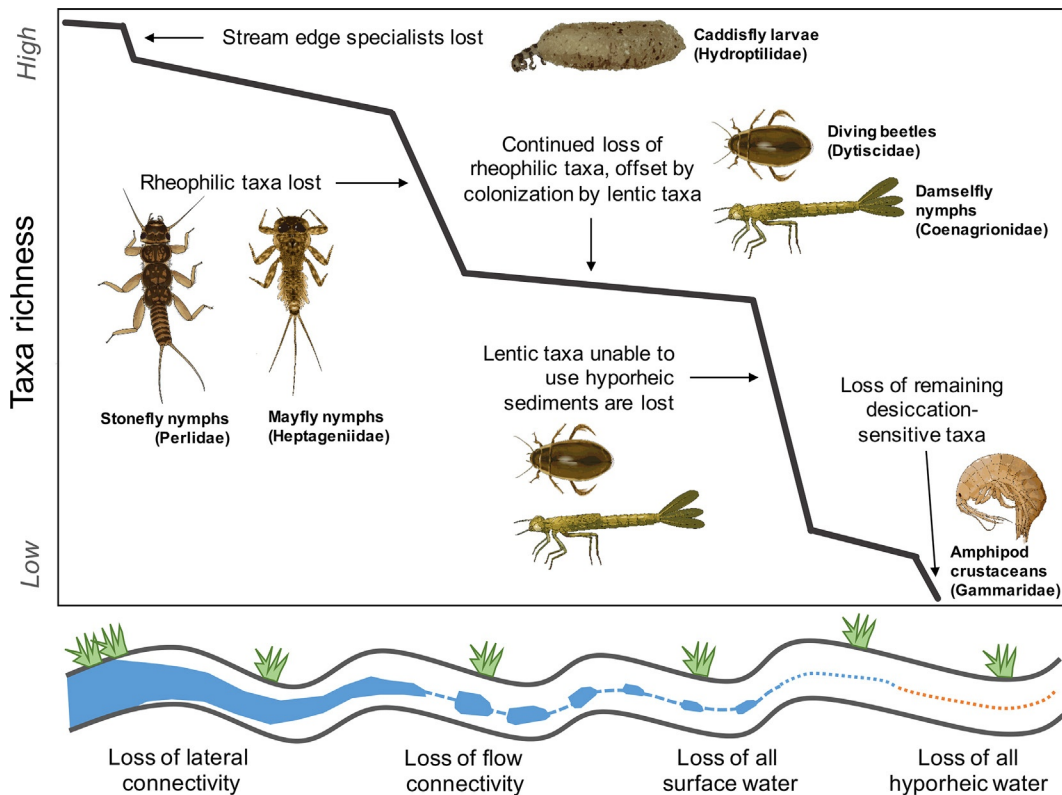


FIG. 4.3.4

The decline in richness with increasing intermittence, indicating critical thresholds in relation to habitat types and availability, as illustrated in the upper panel (adapted from Boulton, 2003).

Invertebrate drawings: Courtesy Pau Fortuño Estrada. Not to scale.

et al., 2014). Taxon-specific oxygen requirements and temperature tolerances are commonly exceeded as pools contract. For example, temperatures in disconnected pools in the temperate Selwyn River, New Zealand, reached 36.6°C (Drummond et al., 2015), higher than the 24-hour tolerance threshold of many aquatic invertebrates (Quinn et al., 1994). If pools dry, complete surface water loss eliminates desiccation-sensitive taxa. This causes steep reductions in taxa richness (Fig. 4.3.4; Boulton and Lake, 2008; Lake, 2011), despite some taxa persisting in refuges (Fig. 4.3.1, Sections 4.3.2 and 4.3.5).

Despite the temporal changes described earlier, and despite interannual variability in rainfall and temperature, total richness can remain stable among years (Bêche et al., 2009; Storey, 2015). In contrast, environmental variability drives differences in community composition among years, in particular between “wet” and “dry” years (i.e., years with different rainfall inputs, resulting in system-specific in-stream conditions ranging from reduced water depth to complete drying; Resh et al., 2013). For example, Bêche and Resh (2007) found that in Mediterranean-climate Californian streams, caddisfly larvae, beetles, and true bugs characterized dry years, whereas several stonefly taxa were associated with wet years. Similarly, in temperate New Zealand IRES, EPT richness was lower while richness of Cladocera and other meiofauna was higher in dry years, due to slower flows (Storey, 2015). Such

interannual variability may have lagged effects, with altered recruitment in one year affecting population densities in subsequent years (Boulton and Lake, 1992).

SPATIAL VARIABILITY IN TAXONOMIC DIVERSITY

Environmental characteristics vary longitudinally in river ecosystems. In response to this environmental variation, alpha diversity of perennial-river aquatic invertebrate communities is predicted to peak in the mid-reaches (Fig. 4.3.5A; Vannote et al., 1980; Chapter 4.9), whereas beta diversity (defined in Box 4.3.1) is typically higher in headwaters due to isolation and habitat diversity (Sánchez-Fernández et al., 2008; Finn et al., 2011; Sarremejane et al., 2017). Whether similar patterns characterize aquatic invertebrate community diversity along IRES is unknown (Datry et al., 2014a). However, patterns may vary depending on the spatial arrangement of intermittent and perennial sections (Fig. 4.3.5), specifically whether intermittent sections occur in headwaters, mid-reaches, or lower reaches; occur as patches; or extend throughout the entire network (Fig. 4.3.5B–E; Datry et al., 2016c; Chapter 2.3).

In IRES with perennial headwaters, richness may decrease with distance downstream if intermittence increases (Fig. 4.3.5B), due to a reduced quantity and quality of food resources (Datry, 2012) as well as increasing distance from upstream recolonists. In contrast, richness may increase from upstream to downstream in catchments with intermittent headwaters, because downstream perennial reaches are refuges that provide recolonists once flow resumes (Fig. 4.3.5C; Sarremejane et al., 2017). Where drying is patchy, richness may remain stable along a river, because dry patches can be recolonized by processes including drift, benthic migrations, and multidirectional dispersal by flying adult insects (Fig. 4.3.5D); the extent of dispersal by flying insects may also influence diversity in systems with intermittent headwaters. Where complete networks dry, richness may decrease with distance upstream if headwaters are isolated and adult insect flight is the dominant recolonization mechanism (Fig. 4.3.5E). Research is needed to confirm these suggested patterns, which are likely to vary depending on local recolonist sources.

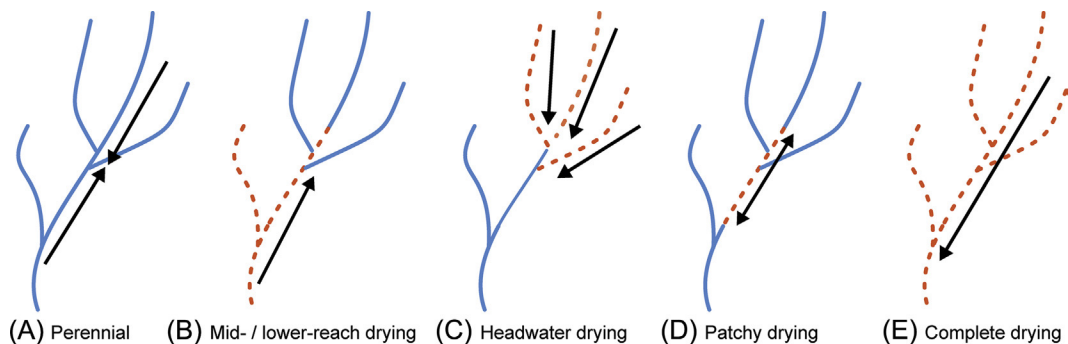


FIG. 4.3.5

Predicted changes in the taxonomic diversity of aquatic invertebrates in a network with different arrangements of perennial (*blue*) and dry (*orange*) reaches. Arrow directions indicate expected increases in alpha diversity according to Vannote et al. (1980) (A) and considering the positions of sources of recolonists following flow resumption (B–E).

As noted by [Dieterich and Anderson \(2000\)](#) for alpha diversity, invertebrate community richness can be higher at intermittent compared to perennial sites due to habitat heterogeneity. In addition, spatial variation in environmental conditions among intermittent sites can mean that spatial beta diversity increases with intermittence ([Schriever and Lytle, 2016](#)). Similarly, in subtropical Australian IRES networks, higher gamma diversity (defined in [Box 4.3.1](#)) at “strongly intermittent” compared to “mildly intermittent” and “near-perennial” sites has been attributed to greater habitat diversity ([Box 4.3.2](#); [Rolls et al., 2016](#)). The few among-region comparisons of IRES richness come from Mediterranean-climate regions. For example, [Bonada et al. \(2008\)](#) found that richness differed among Californian, South African, southwestern Australian, and Mediterranean Basin regions: northern hemisphere regions had higher gamma diversity, probably due to past geological events such as glaciations promoting evolution of new species.

PHYLOGENETIC DIVERSITY

Although [Williams \(1996\)](#) highlighted the wide phylogenetic separation of taxa present in temporary freshwaters, little research has examined the phylogenetic diversity or taxonomic distinctness (defined in [Box 4.3.1](#)) of IRES communities. Some evidence from terrestrial ecosystems suggests that phylogenetic diversity may increase along disturbance gradients ([Giehl and Jaremkow, 2015](#)), whereas the phylogenetic diversity of marine macroinvertebrate communities may decline as disturbance increases due to species-poor higher taxa being lost before those with many representatives ([Clarke and Warwick, 2001](#)). Taxonomic distinctness can also characterize disturbance impacts on community composition ([Marchant, 2007](#)) and may therefore differ between IRES and perennial-river invertebrate communities. Patterns require confirmation for lotic systems including IRES ([Campbell et al., 2008](#)).

4.3.4 FUNCTIONAL DIVERSITY OF IRES INVERTEBRATE COMMUNITIES

The functional diversity (defined in [Box 4.3.1](#)) of IRES invertebrate communities may differ from that in perennial rivers due to greater flow variability and associated changes in resource and habitat availability ([Bêche et al., 2006](#); [Bogan and Lytle, 2007](#)). Taxon loss from IRES communities as drying progresses can reduce functional diversity during certain phases (e.g., after flow ceases and following surface water loss) if the lost taxa have trait states such as rheophily and filter feeding that are not possessed by taxa that remain. The functional diversity of IRES communities may also be lower than that of perennial river communities due to a reduction in large-bodied and long-lived taxa, *K*-selected taxa (which invest heavily in a few offspring), predators, and primary consumers ([Ledger et al., 2013](#); [Schriever et al., 2015](#)). In contrast to these observations, [Leigh et al. \(2016b\)](#) synthesized studies conducted at small spatial scales (i.e., habitats, reaches, or streams) from temperate, Mediterranean, and arid regions and found no differences in functional diversity, richness, or redundancy (defined later) between intermittent and perennial systems among regions. Functional diversity may even be higher in IRES than in perennial streams when multiple seasons are considered, due to the contrasting functional traits of taxa associated with different hydrological phases ([Fig. 4.3.3](#); [Bonada et al., 2007b](#)).

Differences in functional diversity between invertebrate communities in IRES and perennial rivers may be less pronounced than differences in taxonomic diversity, due to functional redundancy among taxa. In IRES, functional redundancy occurs when different taxa perform similar functions but

have contrasting sensitivity to intermittence (Rosenfeld, 2002; Boersma et al., 2014). This redundancy seems most apparent in environmentally harsh systems, which may reflect the prevalence of taxa with traits that promote resistance and resilience to disturbance (Vander Vorste et al., 2016a). For example, Boersma et al. (2014) used experimental pools with different water depths to replicate different phases of the drying process and recorded considerable overlap in the traits of dryland IRES invertebrate assemblages, including body size, feeding mode, and respiration mode. Similarly, Vander Vorste et al. (2016b) reported stable functional diversity despite decreased taxa richness following drying in harsh, braided rivers.

The changes in invertebrate community composition that accompany transitions between flowing, pool, and dry conditions in IRES (Section 4.3.2) can alter functional diversity. For example, Bêche et al. (2006) found that 39 of 73 macroinvertebrate trait states varied between flowing phases and pool/dry phases in the intermittent headwaters of a Californian stream, whereas no traits differed between seasons in a perennial river. Traits that varied in the intermittent stream were associated with survival during dry phases, such as dormancy and desiccation-tolerant life stages (Section 4.3.5), and were more common in the dry season (Bêche et al., 2006).

Feeding groups can be an informative aspect of functional diversity in IRES, with dominant groups changing as drying proceeds (Bogan and Lytle, 2007). The proportion of predators typically increases as flow ceases and prey become concentrated in contracting pools, with beetles and true bugs sometimes accounting for >75% of invertebrates in a “predator soup” (Bogan and Lytle, 2007; Boulton and Lake, 2008). These increases are accompanied by corresponding declines in feeding modes associated with flowing phases, including filter feeding and shredding (Bogan and Lytle, 2007). In contrast, predators are scarce when flow first resumes, while collector-gatherer and scraper/grazer taxa may thrive, feeding on the biofilm-coated detritus that has accumulated on the bed during pool and dry phases (Boulton and Lake, 1992; Bogan et al., 2013). The early stages soon after flow resumption also allow “pioneer” taxa to complete their life cycles relatively free of competition with later colonizers (Langton and Casas, 1999; Williams, 2006). In particular, many dryland streams have distinct pioneer assemblages dominated by winter stonefly, nonbiting midge, blackfly, and dobsonfly juveniles after flow resumes (Fig. 4.3.3; Bogan et al., 2013).

4.3.5 INVERTEBRATE ADAPTATIONS TO FLOW INTERMITTENCE

Adaptations are characteristics that promote an organism’s survival in the prevailing environmental conditions, and include behaviors, physiological and morphological characteristics, and life history strategies (Lytle and Poff, 2004; Chapter 4.10). Each taxon that persists in IRES must possess adaptations that interact to promote resistance (the ability to tolerate conditions *in situ*) and/or resilience (the capacity to recolonize after flow resumes) to flow intermittence (Nimmo et al., 2015; Chapters 4.8 and 4.10). Adaptations typically result in high aquatic invertebrate community resistance in IRES that retain pools (Boersma et al., 2014; Verdonshot et al., 2015), whereas resilience traits are the main mechanisms promoting persistence in IRES that dry (Chester and Robson, 2011; Datry et al., 2014b).

Many aquatic invertebrates in IRES are generalists which have broad environmental preferences and which also inhabit perennial rivers. Some major orders present in IRES include many generalists and fewer specialists. For example, aquatic juveniles of most EPT species are associated with perennial flow (Fig. 4.3.2; Bonada et al., 2007a, 2008), but a few are IRES specialists, including the caddisfly

Mesophylax aspersus and the stonefly *Mesocapnia arizonensis* (Salavert et al., 2008; Bogan et al., 2013). Adaptation to flow intermittence therefore varies at multiple taxonomic levels and may also differ among populations within a species (Lytle et al., 2008a; Chapter 4.10).

REFUGE USE PROMOTES PERSISTENCE IN IRES

Reaches with perennial flow are refuges that typically support many taxa associated with flowing phases in IRES (Boulton, 1989; Chester and Robson, 2011). Flowing upstream refuges allow energy-efficient recolonization of downstream intermittent reaches by taxa that actively enter the drift to disperse to new areas (Fig. 4.3.5B; Brittain and Eikeland, 1988; Paltridge et al., 1997), and downstream drifters may initially dominate assemblages in newly flowing reaches (Townsend and Hildrew, 1976; but see Vander Vorste et al., 2016a). Although many invertebrates may persist in upstream perennial reaches by chance and then drift passively to intermittent downstream reaches, active movement into upstream refuges requires an initial, energetically expensive migration chasing a receding water line (Table 4.3.1; Lytle et al., 2008b; Chapter 4.8). Equally, Boersma and Lytle (2014) followed a giant belostomatid water bug crawling downstream in a dry channel, their observations indicating that drying can initiate refuge-seeking migrations over land. Perennial refuges downstream of intermittent reaches allow taxa with strong swimming ability and a tendency to move into the current (e.g., gammarid amphipods) to recolonize by active upstream movement after flow returns (Fig. 4.3.5C, Table 4.3.1; Stanley et al., 1994). That both downstream and upstream migrations contribute to community recovery is evidenced by comparable resilience in IRES with and without perennial headwaters (Fig. 4.3.5B and C) within months of flow resuming (Datry et al., 2014b), and by genetically comparable populations in adjacent perennial and intermittent reaches (Zickovich and Bohonak, 2007).

Contracting pools are effective refuges for IRES invertebrates that can thrive in these transitional habitats despite declining water quality and lack of flow (Fig. 4.3.1). For example, Dytiscidae diving beetles and Belostomatidae water bugs are air breathers (Table 4.3.1) that surface to replenish oxygen stores (Lake, 2011; Bogan et al., 2015). Some copepods and ostracods can tolerate low oxygen concentrations and can therefore achieve high densities in pools (Storey and Quinn, 2008). Higher taxa typically associated with well-oxygenated water may also have some species that persist in pools, such as *Gumaga* caddisfly larvae that become sedentary to reduce their oxygen requirements (Fig. 4.3.6a; Jackson et al., 1999). If the refuge capacity of diminishing pools declines due to poor water quality or the consumption of finite prey resources, aquatic insects with flying adult life stages (notably true flies and beetles; Bogan and Boersma, 2012) or that crawl in terrestrial habitats (e.g., giant water bugs; Boersma and Lytle, 2014) may emigrate in search of new refuges.

Where perennial surface-water refuges are unavailable, desiccation-sensitive taxa may respond to surface water loss by moving into the wet subsurface habitat of the hyporheic zone (Fig. 4.3.1; Stubbington, 2012). Adaptations to use the hyporheic refuge include active migration as the water table falls and an appropriate morphology: elongate true fly larvae, robust amphipods, and small microcrustaceans are all common subsurface refugees (Table 4.3.1). In addition, smaller individuals of any taxon may be particularly well adapted to hyporheic refuge use (Table 4.3.1). For example, Vander Vorste et al. (2016a) found that invertebrates recolonizing the channel from subsurface sediments were smaller than those present before drying. To persist in subsurface sediments, invertebrates must also be physiologically tolerant of interstitial water quality, including dissolved oxygen concentrations that typically decline with increasing depth below the streambed.

Table 4.3.1 Aquatic invertebrate trait states that promote resistance and/or resilience to intermittence

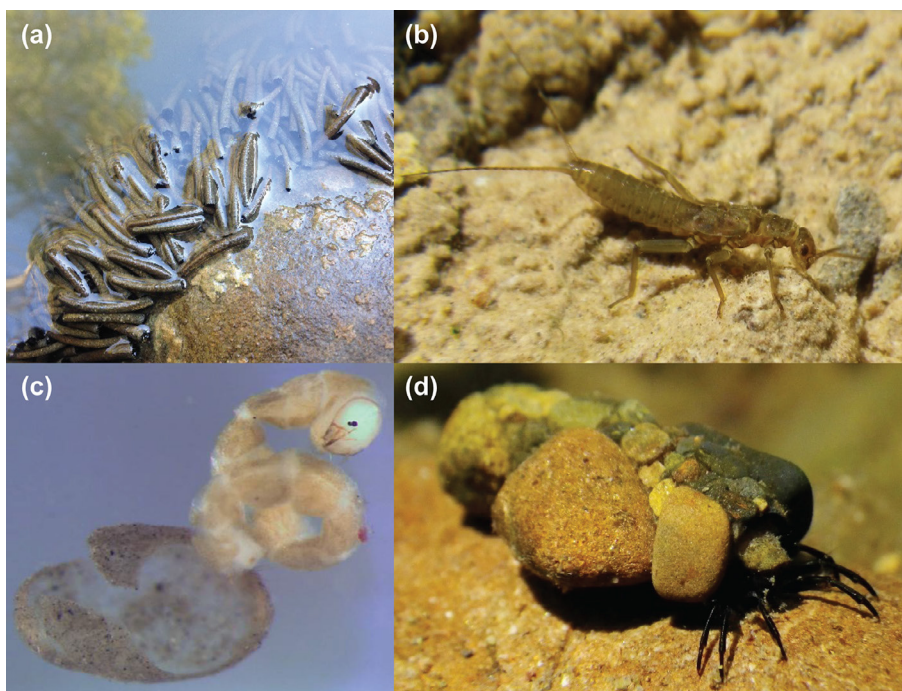
Resistance traits	Trait state	Example taxa	Justification
Body armoring ^a	Strong	Cased caddisfly larvae, e.g., Limnephilidae	Limits water loss from body during dry phases
Desiccation-tolerant life stages ^{a,b}	Present	Stonefly nymphs, e.g., Capniidae and eggs, e.g., Nemouridae	Promotes survival during dry phases
Maximum size ^b	<5 mm	True fly larvae, e.g., Chironomidae	Promotes movement into the hyporheic zone during drying
Respiration ^{a,b}	Air breathers (spiracle/plastron)	Adult true bugs, e.g., Belostomatidae	Enhances survival during dry phases and in low-oxygen pools
Rheophily	Low	Adult beetles, e.g., Dytiscidae	Enhances survival during pool phases
Resilience traits	Trait state	Example taxa	Justification
Adult flying ability	Strong	Caddisfly adults, e.g., Leptoceridae	Facilitates recolonization over larger areas
Adult life span ^a	Long (>1 month)	Diving beetles, e.g., Dytiscidae	Adult females more likely to survive long enough to lay eggs when flow resumes
Female dispersal ^a	High	Dragonfly adults, e.g., Cordulegastridae	Facilitates recolonization from distant perennial refuges
Life cycle duration ^b	Short (≤1 year)	True fly larvae, e.g., Chironomidae	Allows life cycle completion and faster recolonization
Occurrence in drift ^a	Common (typically observed)	True fly larvae, e.g., Chironomidae	Facilitates recolonization from upstream perennial reaches
Maximum crawling rate ^a	High	Crustaceans, e.g., Gammaridae	Promotes movement into the hyporheic zone during drying
		Adult true bugs, e.g., Belostomatidae	Facilitates entrance into upstream and downstream refuges
Reproduction ^b	Giving birth to live young	Crustaceans, e.g., Gammaridae	Allows faster recolonization and population reestablishment
	Terrestrial egg clutches	Beetles, e.g., Hydraenidae	Allows avoidance of dry phases by aquatic life stages
	Asexual reproduction	Worms, e.g., Naididae	Allows faster recolonization and population reestablishment
Relationship to substrate ^b	Inhabiting subsurface interstices	Groundwater crustaceans, e.g., Niphargidae	Survival in saturated interstices after surface water loss
Swimming ability ^a	Strong	Crustaceans, e.g., Gammaridae	Facilitates recolonization from upstream and downstream reaches

Trait states adapted from footnotes “a” and “b” with trait categories translated in Charvet et al. (2000).

Traits and their justification were informed by Bonada et al. (2007a), Robertson and Wood (2010), Robson et al. (2011), and Datry et al. (2014a)

^aPoff et al. (2006).

^bTachet et al. (2002).

**FIG. 4.3.6**

Invertebrates with adaptations promoting persistence in IRES: (a) sedentary *Gumaga* caddisfly larvae congregate in pools; (b) the stonefly *Mesocapnia arizonensis* enters dormancy as nymphs; (c) dormant *Hydrobaenus* chironomid larvae inhabit protective cases; (d) Limnephilidae caddisfly larvae enter dormancy in humidity-trapping cases.

Photos: Courtesy M. Bogan (a, b, and d) and R. Vander Vorste (c).

If benthic and hyporheic sediments dry, only taxa with desiccation-resistant life stages can persist *in situ* (Fig. 4.3.1; Datry, 2012). Few taxa are physiologically adapted to tolerate complete drying (Stanley et al., 1994; Alpert, 2006); exceptions include some microcrustaceans and true flies (Strachan et al., 2015). Notably, larvae of the African “sleeping chironomid” *Polypedium vanderplanki* can survive drying to a moisture content of <8% in a desiccation-tolerant state in which metabolism virtually ceases (Hinton, 1960). Such “anhydrobiosis” is far more common in meiofauna than macroinvertebrates. For example, many tardigrades and nematodes can become metabolically inactive at any life stage (Watanabe, 2006).

Where interstices in a “dry” bed remain humid, taxa with partial desiccation tolerance can persist in the invertebrate “seedbank” (Table 4.3.1; *sensu* Stubbington and Datry, 2013). For example, Datry et al. (2012) recorded 46 aquatic taxa in a temperate IRES seedbank, including beetles, caddisfly and true fly larvae, and many meiofauna. Some of these taxa can survive long periods in moist sediments, notably meiofauna including rotifers and nematodes which remain viable for decades as dormant eggs or developed individuals (Pennak, 1989). Macroinvertebrate traits enhancing physiological desiccation tolerance include body armoring; for example, case-building Limnephilidae caddisfly larvae can persist in interstices for

weeks within humidity-trapping cases (Fig. 4.3.6d, Table 4.3.1; Ruiz-García and Ferreras-Romero, 2007). However, the greater physiological demands of surviving in drier sediments (Wickson et al., 2012) mean that seedbanks may become graveyards as aridity increases (Stubbington and Datry, 2013). Only a few specialists such as dormant *M. arizonensis* stonefly nymphs (Bogan et al., 2015) (Fig. 4.3.6b, Table 4.3.1) and *Hydrobaenus* nonbiting midge larvae (Fig. 4.3.6c) persist in dryland IRES seedbanks.

LIFE CYCLE ADAPTATIONS AND REFUGE USE INTERACT TO PROMOTE SURVIVAL

IRES insects may have life cycle adaptations that involve synchronizing aquatic life stages with flowing phases, and terrestrial adults and dormant eggs with dry phases, to promote persistence in systems with temporally predictable dry phases. For example, desiccation-resistant eggs of the IRES-associated caddisfly *Stenophylax permistus* hatch following flow resumption (Ruiz-García and Ferreras-Romero, 2007). Larvae then develop and pupate during the flowing phase, and adults emerge before the stream dries and spend the dry phase in terrestrial environments (Crichton, 1971). Riparian and terrestrial habitats can therefore provide refuges that promote insect persistence in IRES (Robson et al., 2011; Chapter 4.8, Section 4.3.7). Life cycle synchronization can also involve varying life stage duration in wet and dry years. For example, semivoltine *Neohermes filicornis* fishfly larvae emerge from IRES earlier in dry years (Cover et al., 2015). Such semivoltine insects (having a life cycle exceeding one year) must persist during dry phases as juveniles, typically in a dormant state. *Guadalgenus franzi* stonefly nymphs and *N. filicornis* larvae are among those thought to burrow into hyporheic sediments and become dormant as juveniles during dry phases in Mediterranean-climate IRES (Agüero-Pelegrín and Ferreras-Romero, 2002; Cover et al., 2015).

Following flow resumption, nearby perennial waters are potential sources of IRES recolonists, notably for insects that complete pupation and emergence during flowing phases then fly overland as aerial adults (Bogan and Boersma, 2012). As flow duration lengthens, the terrestrial adults of an increasing number of taxa with desiccation-sensitive aquatic juveniles (e.g., mayflies and dragonflies) have time to recolonize IRES by flight from perennial waters, until community composition comes to resemble its predrying state (Bogan et al., 2015). High female dispersal capacity is therefore a trait associated with resilient IRES insects (Table 4.3.1), as reflected by extensive gene flow between populations and a lack of genetic structure in taxa such as the Australian caddisfly *Lectrides varians* (Wickson et al., 2014; Chester et al., 2015). Genetic analyses also provide evidence of dispersal by flight between isolated IRES, for example, in mayflies and caddisflies (Chester et al., 2015).

Aerial adult insects capable of overland flight recolonize IRES by laying eggs on either water or dry sediments. Eggs deposited on water may develop immediately or may not hatch until triggered by cues such as low temperatures (Hynes, 1976). Such delayed hatching increases the likelihood of aquatic life stages completing development during an uninterrupted flowing phase. Additional “bet-hedging” strategies (Lytle and Poff, 2004), including asynchronous egg hatching (Sandberg and Stewart, 2004), can also increase the chance of at least a partial cohort developing while immersed, despite “false starts” (short periods of immersion; Strachan et al., 2016). Following egg hatching, rapid development facilitates completion of aquatic life stages. For example, chironomid adults may emerge 19 days after larval development begins (Table 4.3.1; Jackson and Sweeney, 1995). For taxa completing their entire life cycle in water, ovoviviparity (giving birth to live young) and asexual reproduction are reproductive traits that facilitate rapid recolonization and population reestablishment following flow resumption in intermittent reaches (Table 4.3.1).

ADAPTATIONS TO IRES ARE TRADE-OFFS THAT ALSO INFLUENCE OTHER ASPECTS OF SURVIVAL

Adaptations that promote persistence during pool and dry phases may increase exposure to other threats; for example, dormant individuals are vulnerable to predation (Robson et al., 2011). Equally, refuge inhabitation may reduce fitness, for example, where drying refuges such as the hyporheic zone have limited food resources (Vander Vorste et al., 2016c). In addition, desiccation tolerance entails metabolic costs that may be offset by slower growth (Alpert, 2006) and by reduced fitness of dormant individuals (Wickson et al., 2014). Invertebrate traits may therefore reflect trade-offs between dry-phase persistence and exposure to other threats, with adaptations differing within and among IRES and climate zones depending on environmental conditions (Welborn et al., 1996), and with genetically distinct populations within a species exhibiting contrasting behavioral responses to different hydrological regimes (Phillipsen and Lytle, 2013). For example, where complete drying occurs, desiccation tolerance may be more common in temperate than dryland fauna, because sediment moisture content remains relatively high and a physiology closer to the “norm” is sufficient for survival (Lytle and Poff, 2004; Stubbington and Datry, 2013). With decreasing humidity, taxon-specific desiccation-tolerance thresholds are exceeded and most dryland IRES taxa instead use resilience traits that facilitate movement into (Stanley et al., 1994; Lytle et al., 2008b) and subsequent recolonization from perennial refuges (Boulton et al., 1992).

The trait combinations that promote persistence in IRES may also enhance survival of other flow extremes. For example, *Gammarus* amphipods may enter the hyporheic zone in response to both increasing and decreasing discharge, and so may be early recolonists after both flooding and drying events (Stubbington, 2012). Equally, physiological adaptations may promote cotolerance of multiple stressors; for example, a strong positive correlation between salinity tolerance and desiccation tolerance has been recorded in beetles from saline IRES (Arribas et al., 2014). Therefore, the trait combinations of IRES invertebrates may not be unique to intermittent systems but reflect the filtering of traits by harsh environmental conditions that can also occur in perennial rivers (Vander Vorste et al., 2016b).

4.3.6 THREATS TO IRES INVERTEBRATE COMMUNITIES

Freshwaters are among the world’s most threatened ecosystems (Strayer and Dudgeon, 2010), and IRES are at particular risk due to their poor public perception (Boulton, 2014; Datry et al., 2014a; Chapters 5.2 and 5.3). Aquatic invertebrates face multiple threats relating to human alteration of the hydrological, physical, chemical, and biological characteristics of IRES (Chapter 5.1). In-stream communities are particularly vulnerable during dry phases, when channels may be viewed as biologically inactive and exploited, for example, for recreation and agriculture (Fig. 4.3.7; Hans et al., 1999; Steward et al., 2012).

Increases in the duration, frequency, and intensity of flow cessation and drying events resulting from climate change and water abstraction severely threaten IRES and their aquatic invertebrates (Chapter 5.1), typically causing alpha and beta diversity to decline as a smaller range of generalist taxa become dominant (Datry et al., 2014b; Section 4.3.3). Equally, artificial perennialization of naturally intermittent systems is altering invertebrate communities such as in dryland and Mediterranean regions where sewage, urban, agricultural, and industrial effluents are released into IRES channels (Múrria et al., 2008; Steward et al., 2012; Chapter 5.1). Artificially perennial flow may be viewed as “improving” ecological quality, but may lead to local extinction of specialist IRES invertebrates which survive because intermittence eliminates their competitors or predators, or because IRES provide conditions that



FIG. 4.3.7

Río Seco (Málaga, Spain) is an intermittent river channel used for recreation during dry phases; a car can be seen driving along the channel.

Photo: Courtesy N. Bonada.

match their functional traits (Section 4.3.4). Loss of intermittence may also threaten taxa associated with isolated pools and dry phases (Box 4.3.2; Section 4.3.3). Other patterns of altered IRES hydrology may be more complex. For example, seasonal and between-year flow variability may be subdued by flow regulation associated with agricultural land use (Belmar et al., 2013); this hydrological homogenization causes community diversity to decline in groups such as aquatic beetles (Bruno et al., 2014).

The consequences of altered flow permanence on aquatic invertebrate biodiversity are poorly understood, but shifts between perennial and intermittent flow may be tipping points that drive communities to alternate states. Artificial intermittence is likely to eliminate taxa from entire networks (Bogan and Lytle, 2011; Datry et al., 2016c), and where lost taxa played key ecological roles, ecosystem processes can be affected. For example, *Gammarus* amphipods release energy for other feeding groups by shredding leaf litter (Section 4.3.1), but are desiccation sensitive (Stubbington et al., 2009b; Vander Vorste et al., 2016c) and their populations decrease as intermittence increases (Datry et al., 2011). Consequently, flow cessation and drying may reduce leaf litter decomposition (Corti et al., 2011), with ecosystem function sometimes remaining altered months after flow resumes (Maamri et al., 1997; Datry et al., 2011). Similarly, net-spinning caddisfly larvae are “ecosystem engineers” that stabilize substrates (Johnson et al., 2009) but may be lost if intermittence increases, which could exacerbate sediment mobilization during floods. Finally, top invertebrate predators can be eliminated by shifts to intermittent flow regimes, triggering cascading effects that extend through food webs and alter community structure (Bogan and Lytle, 2011).

Aquatic invertebrates are also threatened by pollutants, including sewage, pesticides, and fine sediment. The negative effects of such pollutants on invertebrate communities may be exacerbated in IRES by

natural transitions between flowing, pool, and dry phases. As flow recedes and pools contract, the capacity to dilute pollutants may decline and salinization (elevation of dissolved inorganic salt concentrations) may increase (Kefford et al., 2016; Chapters 3.1 and 3.2) compared to perennial streams. In contrast, the natural salinity of some Mediterranean IRES has been diluted by freshwater inputs from irrigated agricultural land, threatening saline-tolerant IRES invertebrates including true fly larvae and beetles (Velasco et al., 2006). Pollutants that accumulate on the bed during dry phases (Hans et al., 1999) are mobilized by flow resumption, with pulses of contaminated water reducing downstream water quality (Datry et al., 2014a; Chapter 3.2); impacts on IRES invertebrates are typically taxon- and contaminant-specific. The negative effects of fine sediment pollution, which include clogging respiratory structures such as gills, may be greater in IRES than in perennial rivers: first, more fine particles are deposited on the bed as flow velocities decline and as flow ceases in IRES; second, blocking of interstitial pathways may prevent invertebrate migrations into the hyporheic zone refuge (Vadher et al., 2015; Section 4.3.5).

Dry riverbeds are sometimes mined for sand and gravel (Chapter 5.1), and this sediment extraction can alter the channel shape and physical characteristics (Fig. 4.3.8), reducing in-stream habitat diversity. Dry-phase assemblages are impacted through removal of desiccation-resistant life stages present in the invertebrate seedbank, which may impair community recovery after flow resumes (Section 4.3.5). In one Bolivian IRES, intensive sediment mining reduced flowing-phase aquatic invertebrate density and richness, and during dry phases, sediments from mining-impacted reaches contained 0%–50% of the taxa found in nonimpacted reaches (T. Datry, personal communication, Fig. 4.3.8).

Outside the channel, clearance of IRES floodplain and riparian zone vegetation for agriculture may occur alongside fertilizer use and water abstraction (Fig. 4.3.9A), with multiple interacting impacts on



FIG. 4.3.8

Rio Biloma (Cochabamba, Bolivia): an intermittent river channel after excavation for sediment mining.

Photo: Courtesy T. Datry.

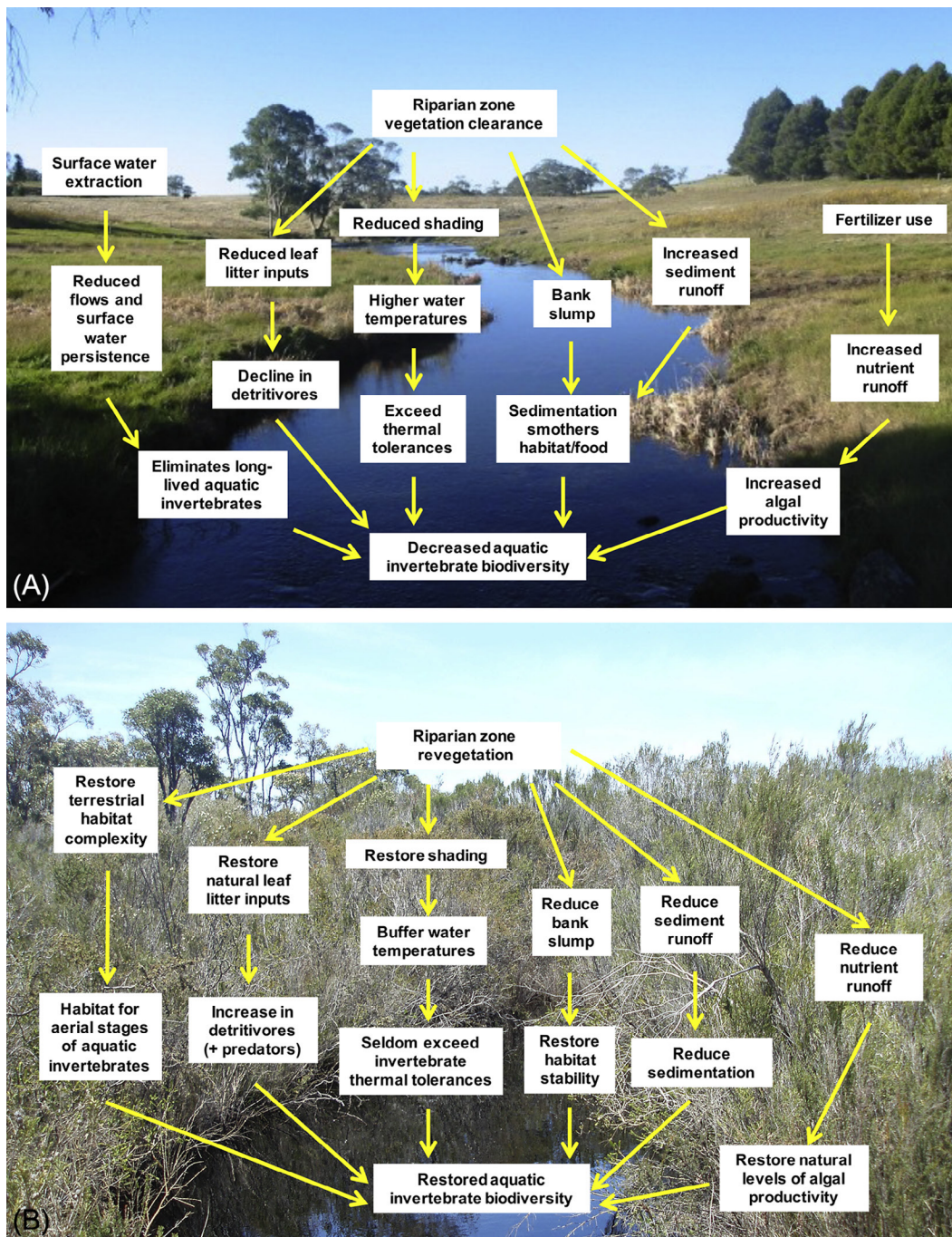


FIG. 4.3.9

Multiple interacting impacts on invertebrate biodiversity resulting from clearance of riparian zone vegetation, fertilizer use and surface water abstraction for agriculture in IRES catchments (A) and the potential reversal of many of these effects by riparian zone restoration (B).

Photos: Courtesy A.J. Boulton.

aquatic invertebrate communities. Reduced vegetation cover can lead to bank erosion and sediment inputs, with fine particles smothering benthic sediments and restricting access to subsurface sediments. Trees, shrubs, and associated habitats, used by terrestrial adult life stages of IRES insects, are removed. Loss of channel shading promotes algal growth in nutrient-rich waters, and water temperatures may exceed the thermal tolerance of some invertebrates, especially when pools contract (Lake, 2011). During dry phases, sediment temperatures increase, potentially exceeding the thermal tolerances of seedbank inhabitants and increasing mortality of desiccation-tolerant life stages (Storey and Quinn, 2013). Finally, loss of riparian vegetation alters the type and amount of organic matter entering IRES channels, shifting aquatic food web structure from dominance by leaf litter shredders to algal herbivores. Collectively, these impacts may reduce aquatic invertebrate diversity (Fig. 4.3.9A).

IRES and their aquatic invertebrate communities are typically exposed to multiple interacting threats because flow regime alteration, pollutant inputs, and land use changes are all concentrated into areas of high human activity (Tockner et al., 2010). For example, groundwater abstraction to irrigate crops is a major cause of artificial intermittence (Vörösmarty and Sahagian, 2000; Chapter 5.1) and is often associated with high pesticide loads, with likely interacting impacts on invertebrate diversity. In addition, altered hydrology may enhance the invasion success of nonnative species. For example, artificial perennialization may facilitate the spread of desiccation-sensitive invasive species such as *Dikerogammarus* “killer shrimps” (Poznańska et al., 2013) and the mudsnail *Potamopyrgus antipodarum* (Múrria et al., 2008). Equally, reduced flow permanence can favor desiccation-tolerant invasive species over desiccation-sensitive natives, as observed for *Orconectes* crayfish (Larson et al., 2009). Novel experiments (e.g., Elbrecht et al., 2016) are needed to examine the cumulative effects of multiple interacting stressors on IRES invertebrates.

4.3.7 MANAGING IRES TO PROMOTE AQUATIC INVERTEBRATE BIODIVERSITY

As described in Section 4.3.6, the main threats to aquatic invertebrate biodiversity in IRES are altered hydrology including changes to flow permanence, deterioration of water quality due to pollution, changes to channel morphology due to sediment extraction, loss or alteration of riparian vegetation, and arrival of invasive taxa (Chapter 5.1). That these threats and their causes typically coincide (Tockner et al., 2010) is highly relevant to ecosystem management because a single, carefully selected management strategy may simultaneously address multiple threats, with widespread benefits for biodiversity.

One management strategy to address multiple threats to IRES invertebrates is to restore native riparian vegetation (Sweeney and Newbold, 2014; Chapter 5.4). Such vegetation stabilizes banks and intercepts runoff containing pesticides, nutrients, fine sediment, and pollutants (Fig. 4.3.9B). Denser shading and decreased nutrient inputs reduce algal productivity, and increased leaf litter inputs may allow shredder taxa to recover their dominance in IRES food webs (Kupilas et al., 2016). Riparian vegetation also provides habitat for the terrestrial adult life stages of insects with aquatic juveniles, and its restoration may promote recolonization of IRES by flying adults following flow resumption (Chester et al., 2015). Provided that recolonization is possible, this management strategy may substantially restore aquatic invertebrate biodiversity (Chapter 4.8), favoring life stages in inundated aquatic habitats as well as aerial adults that use the terrestrial habitat provided by riparian vegetation (Fig. 4.3.9B).

There are, however, few documented cases of IRES restoration (Chapter 5.4) and, seemingly, none published that have specifically targeted invertebrate biodiversity. In contrast to the widespread recognition of the conservation value of temporary pond biota (Van den Broek et al., 2015), equivalent protection of IRES faunas is minimal. Very few aquatic invertebrate species that are solely or largely restricted to IRES are listed for protection, contrasting with the far greater number of species in perennial rivers. One rare example is the snail *Glacidorbis occidentalis*, which is largely restricted to IRES in native eucalypt (jarrah) forest in Western Australia (Bunn et al., 1989), and is listed as Vulnerable on the IUCN Red List of Threatened Species (IUCN, 2015). At a national level, the mayfly *Paraleptophlebia weneri*, which typically inhabits the “winterbourne” headwaters of chalk streams in southern England, is listed as Nationally Scarce (Macadam, 2016). The lack of protection for other rare and potentially threatened aquatic IRES invertebrates probably reflects our limited knowledge of their distribution and ecology, rather than an absence of species needing protection. Regional surveys of temporary ponds (e.g., Karaouzas et al., 2015) have indicated that many rare taxa are restricted to only a few ponds, and protecting habitat diversity and connectivity is crucial for aquatic invertebrate conservation in pond networks (Florencio et al., 2014). Regional conservation strategies are therefore vital to protect aquatic invertebrate diversity in temporary pond networks and the same may be true in IRES.

Crucially, although perennial rivers often support higher aquatic invertebrate diversity during flowing phases (Section 4.3.3), several studies have demonstrated that intermittent reaches support unique, endemic, or locally rare species, contributing to regional-scale (gamma) diversity and warranting inclusion in conservation planning (Sánchez-Fernández et al., 2008; Bogan et al., 2013). Protection measures such as conserving natural flow regimes, riparian vegetation (Fig. 4.3.9B), and perennial refuges have been advocated for maintaining aquatic invertebrate diversity (Chester and Robson, 2011; Cañedo-Argüelles et al., 2015). However, effective management is hampered by limited understanding of the ecology of IRES invertebrates, especially rare species. In addition, management and conservation strategies must be implemented at suitably broad spatial scales that both acknowledge the importance of dispersal dynamics and preserve the natural landscape fragmentation that hinders dispersal of the competitors and predators that exclude certain IRES invertebrates from perennial rivers.

Phylogenetic as well as taxonomic diversity should be considered when identifying nature conservation priorities, as priorities identified using these two approaches can differ (Vane-Wright et al., 1991). In addition, conservation efforts should recognize and protect genetic variability within and among IRES invertebrate taxa. Species with limited dispersal abilities (e.g., the giant water bug *Abedus herberti* and *Geocharax* crayfish) may have genetically distinct populations even at small spatial scales (Phillipsen and Lytle, 2013; Chester et al., 2015), and species with strong dispersal (e.g., the caddisfly *Lectrides varians*) may also be paraphyletic (i.e., comprising multiple distinct genetic lineages; Wickson et al., 2014). Such genetic distinctions among populations within a putative species sometimes warrant taxonomic revisions, and reclassification may increase the rarity of the resultant new species (Wickson et al., 2014).

4.3.8 CONCLUSIONS AND FUTURE RESEARCH PRIORITIES

With climate change predictions of intensifying droughts interacting with increasing water resource pressures and land-use alteration in many regions, flow regimes are changing and intermittence may be increasing (Chapter 1). Consequently, IRES invertebrates face increasing and unpredictable

challenges. Adaptation to flow intermittence typically necessitates adaptation to extremity and unpredictability, and IRES taxa may therefore survive some environmental change. However, biotic responses will vary among and within taxa, and populations are facing pressures to adapt at rates that may prove unachievable. In particular, where the spatial extent of drying increases, genetic bottlenecks stemming from increasing isolation may reduce the ability of populations to adapt, particularly for taxa with limited dispersal (Zickovich and Bohonak, 2007). Restoration initiatives that address multiple interacting stressors are therefore needed to safeguard IRES invertebrate communities. Regulatory agencies need to incorporate IRES and their aquatic invertebrates into long-term, routine biomonitoring programs to ensure changes in community health are documented (Cid et al., 2016). Indices developed for perennial streams are potentially appropriate for such biomonitoring (Mazor et al., 2014), particularly in systems with longer flowing phases (Prat et al., 2014). Finally, the conservation status of specialist IRES invertebrates requires examination at local to international scales, and at genetic to metapopulation levels, to identify populations and species requiring targeted conservation measures.

With IRES becoming an increasingly dominant feature of many river networks, aquatic invertebrates are effective study organisms for addressing wider IRES research challenges, due to their abundance, diversity, and ubiquity. Research priorities that invertebrates could be used to address include exploration of the pathways linking IRES to their catchments and connecting neighboring streams, the long-term (e.g., decadal) temporal dynamism of IRES habitats and communities, and community responses to multiple interacting human activities (Leigh et al., 2016a). Further research is also needed to characterize the contributions that aquatic invertebrates make to IRES biodiversity through their genetic, taxonomic, and phylogenetic diversity, and to highlight the importance of IRES invertebrates in relation to the ecosystem functions they provide.

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